Assessment of Public Satisfaction with Healthcare Across Indian States

DS3006 Advanced Statistics — Final Report, Spring 2026

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Code, data, and supplementary materials: https://github.com/subhisabharwal/pulbic_health_assessment

Abstract

This research analyses public satisfaction with healthcare services across Indian states using the Citizens Survey 2022–23(Harvard Dataverse, 2026), a nationally representative sample of nearly 50,000 households. One challenge in this analysis is the large disparity in the sample sizes across states, which can skew comparisons. We tackle this issue by developing a composite index and adopting a hybrid approach of classical statistical analysis and Bayesian hierarchical modelling. The former methods detect differences and factors affecting satisfaction, while the latter achieves partial pooling, enhancing the accuracy of estimates in low-sample states. Our results show significant regional variations in healthcare satisfaction, and socioeconomic and access-related factors are significant determinants of satisfaction. Bayesian estimates have lower variance and greater stability for smaller states than classical estimates. This research provides a robust approach to assessing healthcare satisfaction and informs policy aimed at reducing regional disparities in India.

1 Introduction

Public satisfaction with health systems is a crucial measure of effectiveness, efficiency, equity and responsiveness in a citizen-centred approach. The Lancet Commission argues that a rights-based healthcare agenda should be focused on patient experience, transparency and inclusivity, making satisfaction a key indicator for assessing policies across states in India.

But there are complexities in analysing large-scale sample surveys like the Citizens Survey 2022-23. The unequal sample sizes of states result in varying degrees of statistical uncertainty, which can be misleading when comparing states (especially those with smaller sample sizes) using simple averages. Furthermore, health-care satisfaction is multi-faceted and can be impacted by structural (access or cost), interpersonal (doctor-patient relationships) and system factors (insurance and infrastructure) that may differ across regions and demographical groups.

This study uses a comprehensive analytic approach, including descriptive statistics, classical inference, and hierarchical Bayesian modeling to address these issues. Hierarchical models enable partial pooling across states, leading to more precise estimates for low-frequency areas while allowing for variation.

This study has three main goals: (i) to compare satisfaction between public and private healthcare facilities, taking into account uncertainty, (ii) to evaluate the effect of health insurance coverage in different settings, and (iii) to identify key factors influencing satisfaction, such as cost, waiting time, quality of care, and rural and urban differences. It further assesses satisfaction with government programs like the Ayushman Bharat scheme, and explores the relationship between satisfaction and utilisation.

Lastly, a sensitivity analysis is conducted to verify the Bayesian estimates under different priors. This study seeks to describe and estimate health care satisfaction in India in a valid and relevant way for policy and practice, and to inform both the methodology of inference and the equity and access of health care.

2 Data Overview

2.1 Source and Acquisition

The data are drawn from the Citizens Survey 2022–23 available on Harvard Dataverse (doi:10.7910/DVN/M1DFBO). The dataset covers households across all states and union territories of India, and includes detailed information on healthcare access, utilisation, expenditure, insurance coverage, and satisfaction with healthcare services at both outpatient (OP) and inpatient (IP) levels.

The data are recorded at the individual (household member) level and include variables spanning demographic and socioeconomic characteristics, geographic identifiers, health status, healthcare utilisation, expenditure, quality of care, insurance coverage, and awareness of public health programs. The main categories of variables are summarised in Table 1.

Table 1: Summary of Key Variable Categories

Category	Key variables
Demographic	Age, gender, education, occupation, religion, social category
Geographic	State, district, rural/urban
Health status	Physical health, mental health
OP care	Facility type, reason, emergency, waiting time, cost, doctor behaviour
IP care	Hospital type, duration, expenditure, staff behaviour
Insurance	Status, type, premium, coverage adequacy
Schemes	Ayushman Bharat enrolment and utilisation
Outcomes	Satisfaction (OP, IP), UHC index score

2.2 Key Variables

Overall satisfaction with healthcare is the primary outcome variable. It is defined as a binary indicator (satisfied vs. not satisfied) constructed from Likert-scale responses aggregated across OP and IP care. In addition, domain-specific satisfaction measures—including waiting time, cost, doctor behaviour, availability of medicines, and cleanliness—are used to analyse the determinants of satisfaction.

The key explanatory variables include geographic identifiers (state and district), rural–urban classification, insurance status and type (Ayushman Bharat, CGHS, ESI, and others), type of healthcare facility (public or private), healthcare expenditure (total and out-of-pocket), accessibility measures (travel time and waiting time), and the Universal Health Coverage (UHC) index.

2.3 Data Preprocessing

2.3.1 Data Loading and Initial Inspection

The dataset was imported from a CSV file using the `pandas` library. Initial inspection was conducted to understand its structure and composition. The dataset dimensions were verified using `df.shape`, while `df.info()` provided details on data types and non-null counts. A comprehensive summary of both numerical and categorical variables was obtained using `df.describe(include='all')`.

To assess representativeness, the distribution of observations across Indian states was examined using `value_counts()` on the `state_name` variable and visualized using a bar chart.

2.3.2 Missing Value Assessment

Missing values were computed for all variables and ranked in descending order using:

```
df.isnull().sum().sort_values(ascending=False)
```

This helped identify columns with high missingness and guided subsequent cleaning steps.

2.3.3 Removal of Sparse Text Columns

Columns containing the substring "oth" (typically "Others, please specify" fields) were removed, as they are often sparsely populated and inconsistent:

```
drop_cols = [col for col in df.columns if 'oth' in col.low
df = df.drop(columns=drop_cols)
```

2.3.4 Threshold-Based Column Filtering

A missingness threshold of 80% was applied. Columns with less than 80% non-null values were dropped to ensure data quality:

```
threshold = 0.8 * len(df)
df = df.dropna(axis=1, thresh=threshold)
```

2.3.5 Feature Selection

A subset of relevant variables was selected based on conceptual importance, including demographics, insurance status, provider type, health status, satisfaction indicators, cost, access, and system-level indices. A cleaned working dataset (`df_clean`) was created for further analysis.

2.3.6 Encoding of Satisfaction Variables

Satisfaction-related variables were originally categorical and were encoded as follows:

- Yes → 1
- No → 0
- CNS / DNA → NaN

The variable `b6v` (partial satisfaction) was encoded as:

- Yes-completely → 1
- Yes-partially → 0.5
- No → 0

All variables were converted to numeric format to ensure consistency.

2.3.7 Composite Satisfaction Score

An overall satisfaction score (`overall_sat`) was computed as the mean of all satisfaction variables:

```
df_clean['overall_sat'] = df_clean[
    sat_cols + ['b6v']].mean(axis=1)
```

This score ranges from 0 to 1, where higher values indicate greater satisfaction.

2.3.8 Insurance Variable Re-encoding

The insurance variable was recoded as:

- Have Insurance → 1
- No → 0

The resulting variable was converted to numeric format for analysis.

3 Exploratory Data Analysis (Using Classical Approach)

3.1 State-wise Sample Size Distribution

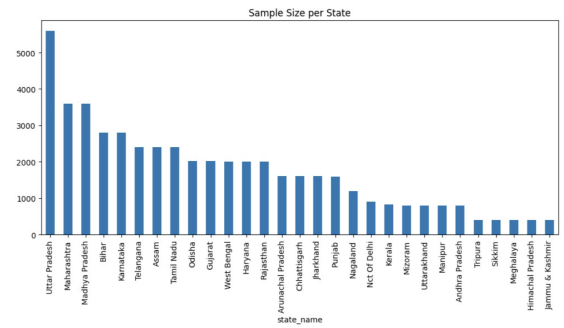

Figure 1: Distribution of sample sizes across states

The distribution of sample sizes across states reveals a significant imbalance, with large states such as Uttar Pradesh contributing several thousand observations, while smaller states such as Sikkim, Tripura, and Jammu & Kashmir have fewer than 500 observations. This indicates a highly skewed dataset in which a few states dominate the sample while many remain underrepresented.

From a statistical perspective, this imbalance affects estimation reliability, as state-level averages derived from smaller samples are subject to higher variance and greater uncertainty. As a result, direct comparisons of classical estimates across states may be unreliable and potentially misleading.

3.2 State-wise Average Satisfaction

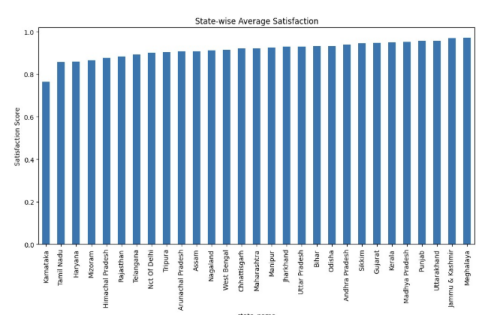

Figure 2: Average satisfaction across states

The state-wise satisfaction scores show limited dispersion, with most states clustering between approx-

imately 0.85 and 0.97, indicating generally high reported satisfaction across the country. The narrow spread makes it difficult to clearly distinguish differences in performance between states, as the values are tightly grouped and lack strong separation. While minor variations are present, the overall clustering suggests broadly similar satisfaction levels, limiting the ability to draw strong comparative insights from the plot alone.

3.3 State-wise Satisfaction with Uncertainty

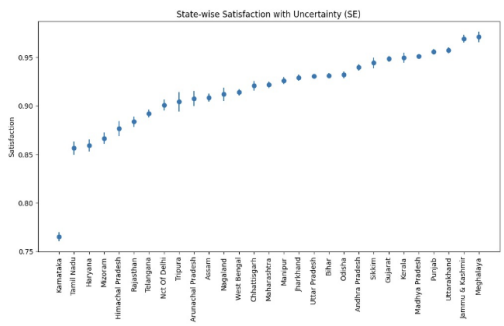

Figure 3: State-wise satisfaction estimates with standard errors

The state-wise satisfaction estimates with associated standard errors highlight substantial variation in the precision of the estimates across states. While the average satisfaction levels appear relatively close across regions, the error bars indicate that states with smaller sample sizes exhibit noticeably higher uncertainty, whereas larger states show tighter confidence bounds.

This demonstrates that differences in point estimates alone may be misleading without accounting for estimation variability. In particular, overlapping uncertainty intervals across many states suggest that apparent differences in satisfaction may not be statistically significant.

3.4 Sample Size and Estimation Uncertainty

The relationship between sample size and standard error exhibits a clear inverse pattern, with larger sample sizes corresponding to lower estimation uncertainty. States with smaller samples display significantly higher standard errors, indicating less precise estimates of satisfaction, while states with larger

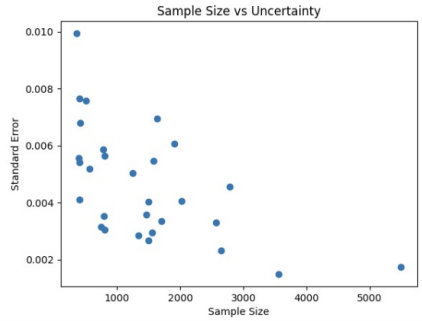

Figure 4: Relationship between sample size and estimation uncertainty

samples show more tightly concentrated and reliable estimates. This pattern reinforces the impact of sample size on the stability and precision of statistical estimates.

3.5 Insurance Status and Satisfaction

The bar plot comparing average satisfaction across regions by insurance status does not reveal any strong or consistent differences between insured and uninsured individuals. The mean satisfaction values for both groups are very similar across most regions, indicating that insurance status does not have a substantial observable effect on satisfaction within the classical framework.

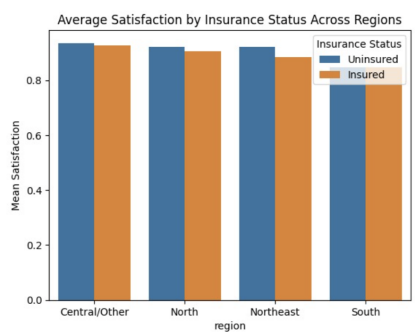

Figure 5: Average satisfaction by insurance status across regions

Furthermore, the variation in average satisfaction across regions is minimal, with values remaining closely clustered. This suggests that regional differences in satisfaction are also limited in magnitude. Based on classical mean comparisons, both insurance status and regional variation exhibit only minor differences in satisfaction levels, making it difficult to

draw strong conclusions from this visualization alone.

3.6 Health Scheme and Satisfaction

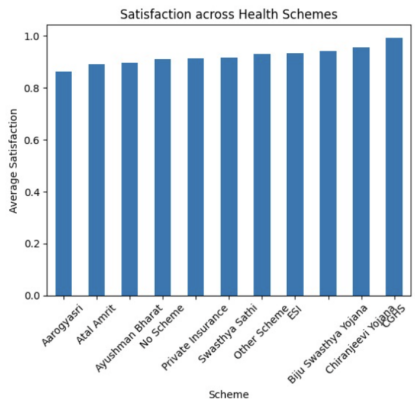

Figure 6: Average satisfaction across different health schemes

The comparison of average satisfaction across different government health schemes does not reveal any substantial variation between groups. The mean satisfaction values for most schemes are closely clustered within a narrow range, indicating that beneficiaries of different schemes report broadly similar levels of satisfaction.

3.7 Public vs Private Healthcare Satisfaction

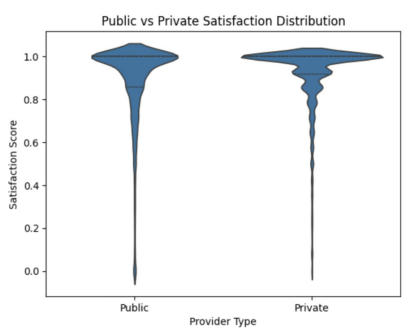

Figure 7: Distribution of satisfaction across public and private healthcare providers

The violin plot of satisfaction across public and private healthcare providers indicates generally high satisfaction levels in both sectors, with values concentrated near the upper end of the scale. The median satisfaction appears similar for both provider types, suggesting no strong difference in overall average experience. However, private healthcare exhibits a slightly tighter distribution, indicating more consistent satisfaction among users. In contrast, public healthcare shows greater variability, with a wider spread and a longer lower tail, suggesting that while many users report high satisfaction, a notable proportion experience comparatively lower satisfaction.

This suggests that, within the classical framework, differences between public and private healthcare are modest and not strongly pronounced in the data.

3.8 Satisfaction Across Gender and Age Groups

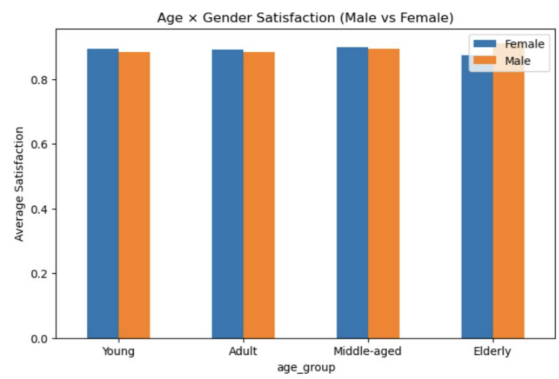

Figure 8: Satisfaction levels across gender and age groups

The chart shows that satisfaction levels are consistently high for both males and females across all age groups, with only minor differences between them. This suggests that gender does not significantly influence satisfaction in most cases. Satisfaction remains stable from young to elderly populations, indicating a broadly uniform experience across age groups for these two categories.

3.9 Rural vs Urban Satisfaction

Satisfaction scores for both rural and urban populations are concentrated near the higher end, indicating generally high satisfaction across both groups. However, rural areas show a sharper peak at the highest values, suggesting more consistently high satisfaction among rural respondents.

In contrast, urban satisfaction is more spread out across mid-to-high values (approximately 0.7–0.9), indicating greater variability in experiences. Over-

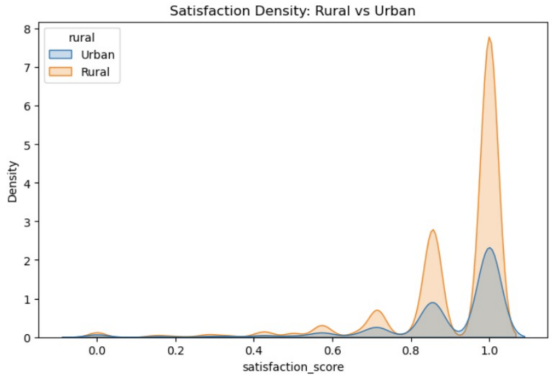

Figure 9: Distribution of satisfaction scores for rural and urban populations

all, while satisfaction is high in both groups, rural responses are more concentrated, whereas urban responses show more variation in perceived service quality.

3.10 Clustering of States by Utilisation and Satisfaction

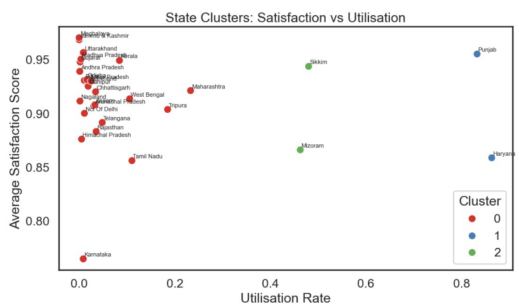

Figure 10: Clustering of states based on healthcare utilisation and satisfaction

The plot identifies three clusters of states based on healthcare utilisation and satisfaction. Most states fall into Cluster 0, characterized by low utilisation but relatively high satisfaction, suggesting that users report positive experiences despite limited use of services.

Cluster 1 (e.g., Punjab and Haryana) shows both high utilisation and high satisfaction, indicating well-performing healthcare systems with good access and quality. Cluster 2 (e.g., Sikkim and Mizoram) reflects moderate utilisation with varying satisfaction levels, suggesting developing systems.

Clustering highlights the clear differences in how states balance healthcare access and service quality.

The exploratory analysis reveals that satisfaction levels are consistently high across states, demographic groups, healthcare schemes, and provider types, with most values concentrated in a narrow range. However, the dataset is highly imbalanced, with large states contributing significantly more observations than smaller ones, leading to substantial differences in estimation precision. States with smaller sample sizes exhibit higher uncertainty, and the widespread overlap in confidence intervals suggests that many apparent differences in satisfaction are not statistically meaningful.

Overall, the combination of limited variation in satisfaction scores and unequal sample sizes restricts the effectiveness of classical comparisons. While minor differences exist (such as slightly higher variability in public healthcare and urban responses), these are not strongly pronounced. This motivates the use of hierarchical Bayesian modeling, which accounts for sampling uncertainty and state-level variation, enabling more stable, reliable, and interpretable estimates of healthcare satisfaction.

4 Methodology

4.1 Classical Approach for Satisfaction Estimation

We initiated our analysis by constructing an overall satisfaction metric using individual-level survey responses. Let S_i denote the satisfaction score for respondent i, computed as the average of multiple sub-domain indicators including waiting time, cost, travel time, doctor interaction, staff behavior, and communication quality.

Formally, the composite satisfaction score is defined as:

$$S_i = \frac{1}{K} \sum_{k=1}^{K} X_{ik} \tag{1}$$

where X_{ik} represents the score for component k, and K is the total number of satisfaction dimensions. This aggregation ensures equal weighting across all sub-domains and provides a unified measure of perceived healthcare quality.

4.1.1 State-wise Aggregation

To examine regional variation, satisfaction scores were aggregated at the state level. Let n_s denote the number of respondents in state s. The mean satisfaction score for each state is computed as:

$$\bar{S}_s = \frac{1}{n_s} \sum_{i \in s} S_i \tag{2}$$

This estimator serves as the primary measure for comparing healthcare satisfaction across states.

4.1.2 Uncertainty Quantification

To account for sampling variability, we computed the standard error of the mean for each state:

$$SE_s = \frac{\sigma_s}{\sqrt{n_s}} \tag{3}$$

where σ_s is the sample standard deviation of satisfaction scores within state s. These standard errors were used to construct confidence intervals and error bars, highlighting the impact of unequal sample sizes across states.

4.1.3 Group-wise Comparisons

In addition to state-level analysis, we conducted subgroup comparisons across demographic categories such as urban and rural populations. Mean satisfaction levels were computed separately for each group:

$$\bar{S}_g = \frac{1}{n_g} \sum_{i \in g} S_i \tag{4}$$

where $g \in \{\text{urban}, \text{rural}\}$. This allowed us to assess disparities in perceived healthcare quality across different population segments.

4.1.4 Hypothesis Testing via ANOVA

To formally examine whether satisfaction differs significantly across states, a one-way Analysis of Variance (ANOVA) was applied. The model is specified as:

$$S_{is} = \mu + \alpha_s + \epsilon_{is} \tag{5}$$

where μ is the overall mean satisfaction, α_s represents the effect of state s, and $\epsilon_{is} \sim \mathcal{N}(0, \sigma^2)$ is the random error term for individual i in state s.

The ANOVA framework tests whether the group means (states) are statistically different. The hypotheses are formulated as:

Null hypothesis (H_0):

$$\mu_1 = \mu_2 = \cdots = \mu_S$$

(indicating no significant difference in mean satisfaction across states)

Alternative hypothesis (H_1):

At least one state mean differs.

4.1.5 Regional and Insurance-based Stratification

To study variation in satisfaction more clearly, states were grouped into regions—North, South, Northeast, and Central/Other. We also considered insurance status, where $j = 1$ denotes insured and $j = 0$ denotes uninsured individuals.

The mean satisfaction for each region and insurance group is computed as:

$$\bar{S}_{r,j} = \frac{1}{n_{r,j}} \sum_{i \in (r,j)} S_i \tag{6}$$

where $n_{r,j}$ is the number of respondents in region r with insurance status j.

This formulation enables comparison of satisfaction across regions as well as between insured and uninsured groups. The results indicate that insured individuals generally exhibit higher satisfaction levels across most regions, and there are observable differences in satisfaction between regions.

4.2 Linear Mixed Effects Model

To address the limitations of simple state-wise averages, we use a Linear Mixed Effects Model (LMM), which accounts for both individual-level factors and variation across states. This approach is particularly useful when sample sizes differ across states.

The satisfaction score is modeled as:

$$S_{ij} = \beta_0 + \beta_1 X_{ij} + u_j + \epsilon_{ij} \tag{7}$$

where i indexes individuals and j indexes states. Here, X_{ij} represents observed variables such as insurance status, residence, and region. The term u_j is a state-specific random effect, and ϵ_{ij} is the individual-level error term.

4.2.1 Fixed and Random Effects

The model includes fixed effects for insurance status, residence (urban/rural), and region, which capture systematic differences in satisfaction.

A random intercept is included for each state to account for unobserved differences between states. This allows each state to have its own baseline satisfaction level.

4.2.2 Partial Pooling

Unlike the classical approach, the LMM applies partial pooling. States with large sample sizes retain their individual estimates, while states with smaller samples are adjusted toward the overall mean.

This results in more stable estimates and reduces the impact of extreme values caused by small sample sizes.

4.2.3 Estimation

The model is estimated using maximum likelihood methods. This involves estimating the fixed effect coefficients along with the variance of the state-level random effects and the residual error.

4.3 Bayesian Approach

4.3.1 Model Motivation

Classical estimates of state-wise satisfaction rely directly on sample means, which can be unreliable when sample sizes vary significantly across states. As observed in the exploratory analysis, smaller states exhibit higher uncertainty, making direct comparisons potentially misleading.

To address this, a Bayesian hierarchical framework is adopted, which allows information to be shared across states and produces more stable estimates through partial pooling.

4.3.2 Model Specification

Let y_i denote the binary satisfaction outcome for individual i, where:

$$y_i \sim \text{Bernoulli}(p_i) \tag{8}$$

The probability of satisfaction is modeled using logistic regression:

$$\begin{aligned}\text{logit}(p_i) = \alpha &+ \beta_1 \cdot \text{Insurance}_i + \beta_2 \cdot \text{Provider}_i \\ &+ \beta_3 \cdot \text{Rural}_i + \beta_4 \cdot \text{Cost}_i + \beta_5 \cdot \text{WaitTime}_i \\ &+ u_{s[i]}\end{aligned} \tag{9}$$

where $u_{s[i]}$ represents the random effect for state s, and

$$u_s \sim \mathcal{N}(0, \sigma^2_{\text{state}}) \tag{10}$$

4.3.3 Prior Specification

Regression coefficients:

$$\beta_j \sim \mathcal{N}(0, \sigma^2) \tag{11}$$

Intercept:

$$\alpha \sim \mathcal{N}(0, \sigma^2) \tag{12}$$

State-level variance:

$$\sigma_{\text{state}} \sim \text{Half-Normal}(0, 1) \tag{13}$$

These priors are weakly informative.

4.3.4 Posterior Inference

Model estimation is performed using MCMC sampling implemented in `PyMC`.

$$p(\theta \mid y) \propto p(y \mid \theta)\, p(\theta) \tag{14}$$

where θ includes all model parameters.

4.3.5 Shrinkage and Partial Pooling

$$\bar{S}_s \mid \theta_s \sim \mathcal{N}(\theta_s, SE_s^2), \quad \theta_s \sim \mathcal{N}(\mu_0, \tau^2) \tag{15}$$

$$E(\theta_s \mid \bar{S}_s) = \frac{\bar{S}_s/SE_s^2 + \mu_0/\tau^2}{1/SE_s^2 + 1/\tau^2} \tag{16}$$

States with small sample sizes undergo stronger shrinkage, improving robustness.

4.3.6 Model Diagnostics and Validation

Model convergence and fit are evaluated using `ArviZ`, including trace plots, posterior distributions, and convergence diagnostics. Posterior predictive checks assess how well the model reproduces observed data patterns.

5 Result Analysis, Findings, Interpretations and Limitations

5.1 Satisfaction Across States

5.1.1 Classical Estimates

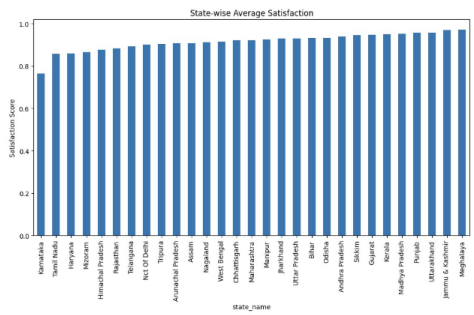

Figure 11: State-wise average satisfaction

The state-wise average satisfaction scores based on classical (raw) means exhibit substantial variation across states, ranging approximately from 0.76 to 0.98. Certain states, such as Meghalaya, Jammu & Kashmir, and Uttarakhand, appear to perform exceptionally well, while others like Karnataka show significantly lower satisfaction levels.

This wide dispersion suggests strong inter-state differences in satisfaction. However, these estimates do not account for differences in sample size or statistical uncertainty, and may therefore exaggerate true variation across states.

5.1.2 Adjusted (Mixed/Hierarchical) Estimates

After applying the adjustment via hierarchical or marginal modelling, the distribution of satisfaction scores becomes noticeably more compressed, with

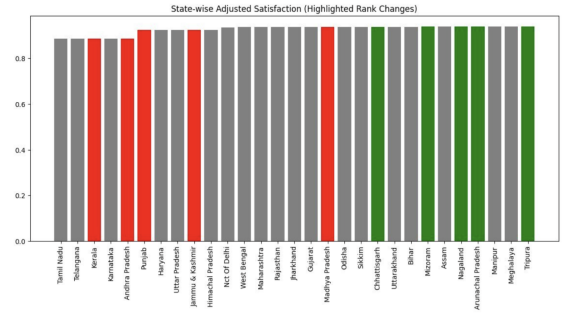

Figure 12: State-wise adjusted satisfaction with highlighted rank changes

most states falling within a narrower range of approximately 0.88 to 0.94.

This indicates that part of the variation observed in the classical estimates was driven by sampling variability rather than genuine differences in satisfaction levels.

Additionally, several states experience changes in their relative rankings:

States highlighted in red show a decline in rank after adjustment, suggesting that their classical estimates were likely overestimated.

States highlighted in green improve in rank, indicating that their initial estimates were relatively underestimated.

5.1.3 Bayesian Adjusted Estimates

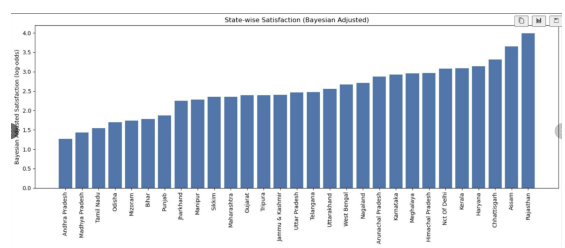

Figure 13: Bayesian-adjusted state-wise satisfaction (log-odds scale)

The Bayesian-adjusted estimates, expressed on the log-odds scale, also demonstrate shrinkage of extreme values relative to the classical results. However, compared to the mixed-effects estimates, the Bayesian approach retains relatively greater variation across states.

This occurs because the Bayesian framework explic-

itly models state-level heterogeneity through random effects, allowing for partial pooling rather than complete smoothing. As a result, while extreme values are moderated, meaningful differences between states are still preserved.

Comparison and Interpretation: A consistent pattern emerges across the three approaches, though the differences are more subtle than extreme. The classical estimates show variation across states, but the dispersion is not excessively wide, indicating that satisfaction levels are generally high and relatively similar across states. However, these estimates are based solely on raw sample averages and do not account for sampling variability, which can affect the reliability of rankings.

The mixed-effects estimates further compress the distribution, producing a more homogeneous set of values. This reflects the partial pooling nature of the method, which smooths out state-level differences and reduces the influence of variability, leading to more stable but less differentiated estimates.

The Bayesian approach provides a balanced adjustment. Through partial pooling, it moderates extreme values while still allowing for meaningful differences across states. Compared to the mixed-effects results, the Bayesian estimates retain slightly greater variation, reflecting state-specific effects, while still improving reliability over the classical approach.

5.2 Posterior Predictive Check

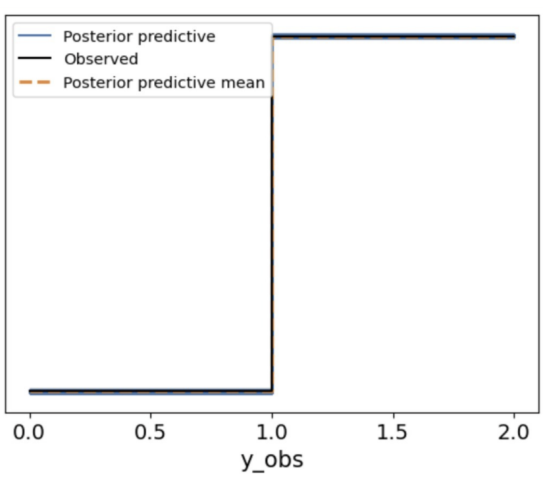

Figure 14: Posterior predictive check for the Bayesian model

The posterior predictive check demonstrates that the model is able to closely replicate the observed data, as the simulated values align closely with the actual outcomes. This indicates that the model provides a strong fit to the data and successfully captures the underlying distribution of the binary response variable.

The step-like pattern reflects the discrete nature of the outcome, while the overlap between observed and predicted values suggests that the model is accurately estimating the probability structure. However, the near-perfect agreement may also indicate limited variability in the outcome or a highly deterministic pattern, which should be interpreted with caution when assessing generalizability.

5.3 Shrinkage: Classical vs Bayesian Estimates

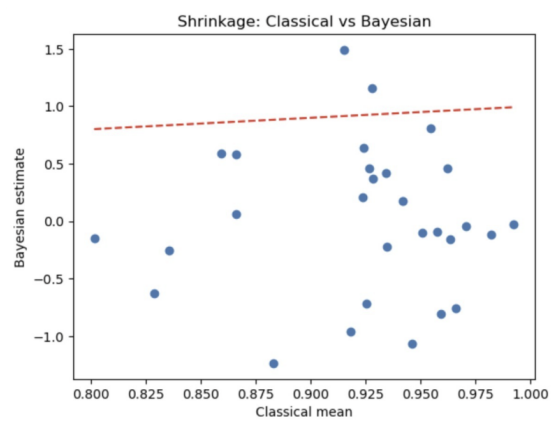

Figure 15: Comparison of classical and Bayesian state-wise estimates illustrating shrinkage

This plot compares classical (frequentist) estimates with Bayesian estimates to illustrate the effect of shrinkage in a hierarchical model. Each point represents a state, where the x-axis shows the classical mean (raw estimate) and the y-axis shows the corresponding Bayesian estimate after adjustment.

A key observation is that the Bayesian estimates are systematically pulled toward a central value, rather than matching the classical estimates exactly. This phenomenon, known as shrinkage or partial pooling, occurs because the Bayesian model incorporates information from the overall population, thereby reducing the influence of noise in individual group estimates.

As a result, extreme classical values—particularly those based on smaller sample sizes—are adjusted toward the global mean, leading to more stable and reliable estimates. The dashed line represents the general trend, indicating that while there is a positive relationship between classical and Bayesian estimates, the Bayesian values are less dispersed.

Overall, this plot highlights how the Bayesian approach mitigates overfitting and variability by balancing individual group information with overall population-level structure.

5.4 Analysis of Variance (ANOVA)

A one-way Analysis of Variance (ANOVA) was conducted to examine whether mean satisfaction levels differ across states.

Hypotheses

Null hypothesis (H_0):

$$\mu_1 = \mu_2 = \cdots = \mu_{29}$$

Alternative hypothesis (H_1): At least one state has a mean satisfaction level that differs from the others.

Results

The ANOVA results indicate a statistically significant effect of state on satisfaction:

$$F(28, 41774) = 127.74, \quad p < 0.001$$

Decision

Since the p-value is less than the significance level ($\alpha = 0.05$), the null hypothesis is rejected.

Interpretation

This result suggests that there are statistically significant differences in mean satisfaction levels across states. However, the between-state variation ($SS = 94.64$) is relatively small compared to the within-state variation ($SS = 1105.37$), indicating that most of the variability in satisfaction occurs within states rather than between them.

Thus, while differences across states are statistically significant, their practical magnitude is relatively modest. This is consistent with earlier findings that satisfaction levels are generally high and clustered within a narrow range across states.

Table 2: ANOVA Results for State-wise Satisfaction

Source	Sum of Squares	df	F	p-value
State	94.643	28	127.741	< 0.001
Residual	1105.367	41774	–	–

5.5 Bayesian Regional Satisfaction

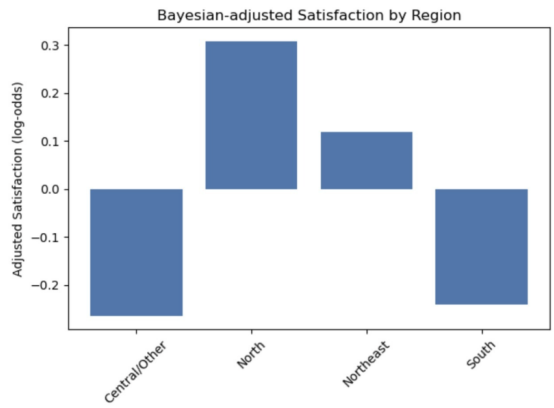

Figure 16: Bayesian-adjusted satisfaction across regions (log-odds scale)

This plot presents the Bayesian-adjusted satisfaction levels across different regions, expressed in log-odds. The results indicate clear regional variation after accounting for model adjustments and controlling for other factors.

The North region shows the highest adjusted satisfaction, suggesting that, on average, it performs better than other regions even after accounting for covariates and sample size differences. The Northeast region also exhibits a positive adjusted satisfaction, though lower than the North, indicating moderately better outcomes relative to the overall average.

In contrast, the South and Central/Other regions display negative adjusted satisfaction values, implying that their outcomes fall below the overall mean after adjustment. These differences suggest that regional factors, possibly including infrastructure, accessibility, or service quality, may influence satisfaction levels.

Overall, the Bayesian adjustment provides a more reliable comparison by reducing the effect of noise and highlighting systematic regional disparities.

6 Limitations

This study is subject to several limitations that should be considered when interpreting the results.

Cross-sectional design. The Citizens Survey 2022–23 is a single cross-section; causal claims about scheme impact or insurance effects cannot be established without longitudinal or quasi-experimental variation.

Subjective satisfaction measures. Satisfaction is self-reported and may reflect expectations rather than objective quality of care. Social desirability bias may inflate reported satisfaction, particularly for government facilities.

Measurement constraints. The constructed satisfaction index relies primarily on outpatient-care variables and may not fully capture all dimensions of healthcare quality, especially for inpatient services and long-term outcomes.

Data quality issues. Scheme participation, including Ayushman Bharat enrolment and utilisation, is self-reported and subject to recall bias. This may introduce measurement error in estimating scheme effects.

Sample size imbalance. State-level sample sizes vary substantially; estimates for highly data-sparse states (e.g., $n < 120$) remain approximate despite hierarchical modelling.

Unobserved confounding. Political, institutional, and regional factors may influence both healthcare provision and satisfaction but are not fully captured in the dataset.

Survey design limitations. Survey weights and complex sampling adjustments are not applied due to missing design variables, which may limit population-level representativeness.

Despite these limitations, the study provides a robust framework for analyzing healthcare satisfaction in India.

7 Future Scope

Several extensions can strengthen and deepen the present analysis.

Longitudinal and causal analysis. Linking the Citizens Survey 2022–23 with datasets such as NFHS or DLHS would enable quasi-experimental designs (e.g., difference-in-differences) to identify causal effects of insurance schemes and healthcare interventions.

Improved data integration. Incorporating survey weights, direct measures of public health expenditure, and district-level infrastructure indicators would enhance representativeness and improve modeling of structural determinants.

Enhanced hierarchical modeling. Extending the Bayesian framework to include district- and region-level random effects, as well as spatial dependence (e.g., CAR priors), would improve estimates for geographically related and data-sparse regions.

Richer satisfaction measurement. Future work could move beyond equal-weight indices by applying factor analysis or weighted scoring methods to better reflect the relative importance of different satisfaction dimensions.

Natural language analysis. Open-ended responses could be analyzed using topic modeling or transformer-based methods to capture qualitative drivers of satisfaction not reflected in structured variables.

Temporal validation. Replicating the analysis on future survey waves would allow assessment of the stability of satisfaction patterns, clusters, and key drivers over time.

8 Conclusion

This study examines public satisfaction with healthcare services across Indian states using a combination of classical statistical analysis, linear mixed-effects modelling, and Bayesian hierarchical methods. Using data from the Citizens Survey 2022–23, the analysis addresses key challenges arising from unequal sample sizes, limited variation in reported satisfaction, and the multidimensional nature of healthcare experiences.

The results indicate that overall satisfaction levels are consistently high across states, demographic groups, and provider types, with most values concentrated in a narrow range. However, this apparent uniformity masks important statistical limitations. Classical comparisons based on mean satisfaction scores reveal only minor differences across states, insurance groups, and regions, and these differences are often not statistically meaningful due to overlapping uncertainty intervals.

A central contribution of this study is the demonstration that state-level comparisons must account for estimation uncertainty. The relationship between sample size and standard error highlights that smaller states exhibit substantially higher variability, making naive comparisons unreliable. The linear mixed-effects model improves estimation by incorporating both individual-level predictors and state-level variation, enabling partial pooling and reducing instability in small-sample states.

The Bayesian hierarchical framework further strengthens the analysis by explicitly modelling uncertainty and applying shrinkage to state-level estimates. This produces more stable and credible estimates, particularly for data-sparse regions, where classical methods tend to overstate differences due to noise.

Substantively, the findings suggest that satisfaction differences across public and private providers, insurance status, and regions are present but modest. Variability within groups often exceeds variability between groups, indicating that perceived healthcare quality is broadly similar across much of the country, with pockets of variation rather than stark divides.

Overall, the study highlights the importance of moving beyond simple averages toward uncertainty-aware and hierarchical modelling approaches when analysing large-scale survey data. By integrating classical and Bayesian methods, the analysis provides a more reliable framework for assessing regional disparities in healthcare satisfaction and offers a stronger foundation for evidence-based policy evaluation.

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